

Question 1:

Aug 2026, 11:59 AM

Level 2 — Medium (M) Matrix Mult. — Array

Write a program for matrix multiplication?

Sample Input:

Mat1 =
1 2
5 3
Mat2 =
2 3
4 1
Sample Output:
Mat Sum =
10 5
22 18

Your Code

```
1 import java.util.*;  
2 public class Main{  
3     public static void main(String[] args){  
4         Scanner sc = new Scanner(System.in);  
5         int[][] a= new int[2][2];  
6         int[][] b=new int[2][2];  
7         int[][] c= new int[2][2];  
8  
9         System.out.println("Enter the matrix1");  
10        for(int i=0;i<2;i++){
```

Output

```
Enter the matrix1  
Enter the matrix2  
10  
5  
22
```

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```
11        for(int j=0;j<2;j++){  
12            a[i][j]=sc.nextInt();  
13        }  
14    }  
15    System.out.println("Enter the matrix2");  
16    for(int i=0;i<2;i++){  
17        for(int j=0;j<2;j++){  
18            b[i][j]=sc.nextInt();  
19        }  
20    }  
21    for(int i=0;i<2;i++){  
22        for(int j=0;j<2;j++){  
23            for(int k=0;k<2;k++){  
24                c[i][j]=c[i][j]+a[i][k]*b[k][j];  
25            }  
26        }  
27    }  
28    for(int i=0;i<2;i++){  
29        for(int j=0;j<2;j++){  
30            System.out.println(c[i][j]+" ");  
31        }  
32    }  
33    System.out.println();  
34 }  
35 }
```

Question2:

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Level 2 — Medium (M) Perfect Num. Gen. — Numbers

Write a program to print the first n perfect numbers. (Hint Perfect number means a positive integer that is equal to the sum of its proper divisors)

Sample Input:
N = 3

Sample Output:
First 3 perfect numbers are: 6 , 28 , 496

Testcases:
1. N = 0
2. N = 5
3. N = -2
4. N = -5
5. N = 0.2

Your Code

```
1 import java.util.Scanner;  
2 public class Main{  
3     public static void main(String[]arg){  
4         Scanner sc=new Scanner(System.in);  
5         int n=sc.nextInt();  
6         System.out.print("first "+n+" perfect numbers are:  
7         int count=0;  
8         int num=1;  
9         while(count<n){
```

Output

```
first 3 perfect numbers are:  
6,28,496
```

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192465061RB

```
8     int num=1;  
9     while(count<n){  
10        int sum=0;  
11        for(int i=1;i<num;i++){  
12            if(num%i==0){  
13                sum+=i;  
14            }  
15        }  
16        if(sum==num){  
17            System.out.print(num);  
18            count++;  
19            if(count<n){  
20                System.out.print(", ");  
21            }  
22        }  
23        num++;  
24    }  
25    System.out.println();  
26 }  
27 }  
28
```

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Question3:

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Level 2 — Medium (M) Bonus — Control ✓ Submitted

In an organization they decide to give bonus to all the employees on New Year. A 5% bonus on salary is given to the grade A workers and 10% bonus on salary to the grade B workers. Write a program to enter the salary and grade of the employee. If the salary of the employee is less than \$10,000 then the employee gets an extra 2% bonus on salary. Calculate the bonus that has to be given to the employee and print the salary that the employee will get.

Sample Input & Output:

Enter the grade of the employee: B
Enter the employee salary: 50000
Salary=50000
Bonus=5000.0
Total to be paid:55000.0

Testcases:

1. Enter the grade of the employee: A
Enter the employee salary: 8000
2. Enter the grade of the employee: C
Enter the employee salary: 60000
3. Enter the grade of the employee: B
Enter the employee salary: 0
4. Enter the grade of the employee: 38000
Enter the employee salary: A

Your Code

```
1 import java.util.*;  
2 public class Main{  
3     public static void main(String[] args){  
4         Scanner sc = new Scanner(System.in);  
5         System.out.println("Enter the grade:");
```

Output

```
Enter the grade:  
enter the salary  
8000.0  
960.0  
8960.0
```

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JAVA PROGRAMMING ANNAMREDDY BHARGAV REDDY 192465061RB

```
6 String grade=sc.next();  
7 char ch =grade.charAt(0);  
8 System.out.println("enter the salary");  
9 double salary=sc.nextInt();  
10  
11 if(ch=='A'){  
12     if(salary>10000){  
13         double bonus=salary*5/100;  
14         System.out.println(salary);  
15         System.out.println(bonus);  
16         double total=salary+bonus;  
17         System.out.println(total);  
18     }  
19     else{  
20         System.out.println(salary);  
21         double bonus=salary*7/100;  
22         System.out.println(bonus);  
23         double total=salary+bonus;  
24         System.out.println(total);  
25     }  
26 }  
27 else if(ch=='B'){  
28     if(salary>10000){  
29         double bonus=salary*10/100;  
30         System.out.println(salary);  
31         System.out.println(bonus);  
32         double total=salary+bonus;  
33         System.out.println(total);  
34     }
```



```
23         System.out.println(bonus);
24         double total=salary+bonus;
25         System.out.println(total);
26     }
27     else if(ch=='B'){
28         if(salary>10000){
29             double bonus=salary*10/100;
30             System.out.println(salary);
31             System.out.println(bonus);
32             double total=salary+bonus;
33             System.out.println(total);
34         }
35     }
36     else{
37         System.out.println(salary);
38         double bonus=salary*12/100;
39         System.out.println(bonus);
40         double total=salary+bonus;
41         System.out.println(total);
42     }
43 }
44 }
```

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Question4

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Level 2 — Medium (M) Count Chars — Strings ✓ Submit

Write a program to read a character until a * is encountered. Also count the number of uppercase, lowercase, and numbers entered by the users.

Sample Input:
Enter * to exit...
Enter any character: W
Enter any character: d
Enter any character: A
Enter any character: G
Enter any character: g
Enter any character: H
Enter any character: *

Sample Output:
Total count of lower case:2
Total count of upper case:4
Total count of numbers -0

Testcases:

1. 1,7,6,9,5
2. S, Q, l, K,7, j, M
3. M, j, l, &, @, G
4. D, K, I, 6, l, *
5. *, K, A, e, l, 8, %, *

Your Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
```

Output

```
enter the String:
Upper cases 1
```



```
4 Scanner sc = new Scanner(System.in);
5 System.out.println("enter the String:");
6 String str= sc.next();
7 int upper=0;
8 int lower=0;
9 int number=0;
10 for(int i=0;i<str.length();i++){
11     char ch=str.charAt(i);
12
13     if(ch==' ')
14         break;
15     if(ch>='A' && ch<='Z'){
16         upper++;
17     }
18     else if(ch>='a' && ch<='z'){
19         lower++;
20     }
21     else if(ch>='0' && ch<='9'){
22         number++;
23     }
24
25 }
26 System.out.println("Upper cases "+upper);
27 System.out.println("lower cases "+lower);
28 System.out.println("number "+number);
29 }
30
31 }
```

number 1